PC Building Empire

Game Design Document (GDD)

1. Executive Summary & Vision

- **Title / Working Name:** PC Building Empire (Tux's Rig Empire)
- **Genre:** 2D Isometric Simulation / Time Management / "Spinning Plates" Tycoon
- **Target Platform:** HTML5 (Browser-native via WebGL / Phaser 3), Desktop packaging via Tauri / Electron
- **Visual Style:** Cute 2D isometric pixel art (similar to Stardew Valley / Overcooked), featuring animated penguin characters
- **Core Philosophy:** A tactile, multi-tasking management loop where manual actions gradually transition into automated logistics, periodically disrupted by high-stakes customer support crises.

2. Controls & Input Architecture

The game supports single-player and couch co-op out of the box with flexible input schemes:

2.1 Control Mapping

- **Player 1 (Lead Penguin):**
  - Movement: `W`, `A`, `S`, `D` (8-directional isometric navigation)
  - Action / Interact: `E`
- **Player 2 (Free Penguin Helper):**
  - Movement: Arrow Keys (`↑`, `↓`, `←`, `→`)
  - Action / Interact: `Numpad 0` (Fallback: `Enter` or `Right Shift`)
  - Upkeep Cost: $0 wages (Player 2 acts as a free, manual human partner)
- **Controller Support (HTML5 Gamepad API):**
  - Full compatibility with Xbox (A to interact), PlayStation (Cross to interact), Steam Controller, Steam Deck, and generic gamepads.
  - Mixed input support: Play with 2 controllers, 1 controller + 1 keyboard, or 2 players on 1 keyboard.
  - Analog stick deadzone filtering (0.2 threshold) and normalized diagonal movement.

2.2 Co-op Interaction Rules

- **Station Mutex:** Each desk or terminal permits only one penguin at a time to prevent race conditions.
- **Item Passing:** Finished boxed PCs can be picked up, dropped, and transferred between players before reaching the dispatch zone.
- **Shared Economy:** Players share a common cash balance, debt liability, and company star rating.

3. Core Micro-Loop (Single PC Lifecycle)

Every computer follows a strict 8-step finite state machine:

1. **Buy Parts:** Player interacts with desk ($1,200 component cost, 1s active click).
2. **Assemble Hardware:** Player holds action key at the desk (4s active hold).
3. **Install Linux OS:** Boot PC and trigger installation (5s passive progress bar; player is free to leave the desk).
4. **List for Sale:** Turn off PC, interact with the Admin Terminal (1s active click).
5. **Buyer Matching:** Background sales algorithm finds a buyer (10s passive countdown).
6. **Box & Label:** Player packs the sold PC into a shipping box and applies a label (3s active hold).
7. **Carry to Dispatch:** Player carries the box to the shipping placement bay (~2s travel time).
8. **Delivery Collection:** Delivery van arrives, loads the PC, and credits cash (15s passive timer).

- **Total Elapsed Time per PC:** ~41 seconds
- **Active Hands-On Time:** ~11 seconds
- **Design Purpose:** Passive timers create natural windows for players to "spin plates" across multiple desks.

4. Economic Balancing & Unit Economics

4.1 Base Unit Economics

- **Component Cost:** $1,200.00
- **Base Sale Price:** $1,320.00 (+10% base margin)
- **Base Net Profit:** $120.00 per PC
- **Starting Capital:** $2,500.00 (ensures working capital buffer for 2 concurrent builds)

4.2 Early Game Walkthrough (Desks 1 to 2)

- PC #1: Spend $1,200 → Sell for $1,320 → Balance: $2,620
- PC #2: Spend $1,200 → Sell for $1,320 → Balance: $2,740
- PC #3: Spend $1,200 → Sell for $1,320 → Balance: $2,860
- **Desk 2 Unlock Cost:** $400.00
- After purchasing Desk 2, player has $2,460 remaining—exactly enough to buy parts for both Desk 1 and Desk 2 simultaneously ($2,400) with a $60 buffer.

4.3 Scale Modifiers (1 to 200 Desks)

- **Component Bulk Discount:** -0.15% per desk (Capped at -30%)
- **Reputational Markup:** +0.20% per desk (Capped at +40%)
- *At Desk 10 (Garage Cap):* Parts $1,182.00 (-1.5%), Sale $1,346.40 (+2.0%), Profit: $164.40/PC
- *At Desk 100 (Warehouse 2):* Parts $1,020.00 (-15.0%), Sale $1,584.00 (+20.0%), Profit: $564.00/PC
- *At Desk 200 (Industrial HQ):* Parts $840.00 (-30.0%), Sale $1,848.00 (+40.0%), Profit: $1,008.00/PC

5. Building Tiers & Capacity Progression

Building Tier	Max Build Desks	Max Helpdesks
Garage	10	1
Warehouse Tier 1	50	5
Warehouse Tier 2	100	10
Industrial Complex	200	20

5.1 Garage Desk Unlocks

- Desk 1: Free
- Desk 2: $400
- Desk 3: $900
- Desk 4: $1,600

- Desk 5: $2,600
- Desk 6: $4,000
- Desk 7: $6,000
- Desk 8: $8,500 (Unlocks Penguin Hiring)
- Desk 9: $12,000
- Desk 10: $16,500 (Unlocks Helpdesk & Station 1)

6. Automation: Penguin Helpers

Unlocked at Desk 8. Each helper automates a specific task in the production pipeline.

6.1 Roles & Wages (120-Second Day Cycle)

- **Buyer Penguin:** Hire: $2,000 | Wage: $150/day — Orders parts for idle desks.
- **Builder Penguin:** Hire: $3,500 | Wage: $250/day — Assembles hardware on desks.
- **Installer Penguin:** Hire: $2,800 | Wage: $200/day — Boots and installs Linux OS.
- **Seller Penguin:** Hire: $2,500 | Wage: $180/day — Lists ready PCs at the Admin Terminal.
- **Dispatcher Penguin:** Hire: $3,200 | Wage: $220/day — Boxes, labels, and carries PCs to dispatch bay.
- **Support Tech Penguin:** Hire: $5,500 | Wage: $400/day — Automatically answers and resolves Helpdesk tickets.

6.2 Payroll & Bankruptcy Risk

Penguin wages are deducted automatically at the end of each 120-second day cycle. Hiring excessive workers without matching desk throughput drains cash and leads to insolvency.

7. The Helpdesk System & Review Ratings

Unlocked at 10 desks. Phone calls and ticket beeps introduce operational friction.

7.1 Mechanics

- **Ticket Trigger:** Random incoming ticket alert with a 10-second countdown to answer.
- **Frequency:** 1 ticket every 75–90s base, scaling with desk count (+1 ticket frequency tier per 10 desks).
- **Resolution Lock:** Requires 20 seconds of continuous progress bar interaction.

7.2 Rating & Penalty Matrix

- **≤ 6s to answer:** 5 Stars → +$200 cash bonus
- **7s to answer:** 4 Stars → +$100 cash bonus
- **8s to answer:** 3 Stars → +$40 cash bonus
- **9s to answer:** 2 Stars → $0 bonus
- **10s (Missed):** 1 Star → -10% sale price penalty on next 3 PCs sold
    - Sale price drops from $1,320 to $1,188 (-$132).
    - Since parts cost $1,200, the player loses -$12 per PC on those 3 sales.
    - Penalties stack: multiple missed calls compound losses, creating a potential bankruptcy spiral.

8. Technical Architecture (HTML5 & WebGL)

- **Rendering:** WebGL 2D Isometric projection using sprite sheets and dynamic Y-sorting (`screenY = (x + y) * (tileHeight / 2)`).
- **AI Pathfinding:** EasyStar.js / A* grid navigation with collision masking around active desks.
- **Input Management:** Unified input merger reading keyboard state map and `navigator.getGamepads()`.
- **State & Economy:** Decoupled fixed-rate simulation tick (20 Hz) with local storage persistence (`localStorage` / `IndexedDB`).
- **Optimization:** Viewport culling and sprite batching to maintain 60 FPS up to 200 desks and dozens of penguins.

9. Graphic Asset Manifest & 2D Isometric Sprite Atlas

9.1 Technical Specifications

- **Perspective:** Standard 2:1 Dimetric/Isometric projection (e.g. 64px width × 32px height floor diamonds).
- **Base Cell Size:** 64 × 64 pixels (uniform grid cell for sorting and sprite slicing).
- **Atlas Texture Dimensions:** 512 × 256 pixels (8 columns × 4 rows of 64×64 frames).
- **Color Palette (16-Bit Nostalgic / Stardew Aesthetic):**
    - Tuxedo Black (`#181425` / `#222034`), Clean White (`#F5F5FA` / `#E6EBF0`)
    - Beak & Feet Orange (`#FF961E` / `#D26414`)
    - Player 1 Red Tie (`#DC323C`), Player 2 Blue Bandana (`#288CF0`)
    - Workbench Wood (`#B48250` / `#8C5F37` / `#644123`)
    - Computer Metal (`#BEC8D2` / `#828C9B` / `#4B505F`)
    - Linux Terminal Glow (`#3CDC78`), Cardboard Box (`#D2AF7D` / `#A58255`)

- Dispatch Hazard Stripes (`#F5CD23` / `#2D2D32`)

9.2 Sprite Atlas Grid Coordinates (512 × 256)

Row 0 (Y: 0 – 63) — Characters & Workforce

- `[0, 0]`: **Player 1 (Tux) Idle** — Red bowtie, centered posture.
- `[1, 0]`: **Player 1 Walk Frame 1** — Right foot forward, flipper sway.
- `[2, 0]`: **Player 1 Walk Frame 2** — Left foot forward, alternate sway.
- `[3, 0]`: **Player 2 (Co-op) Idle** — Blue bandana, distinct nametag slot.
- `[4, 0]`: **Player 2 Walk Frame** — Co-op helper walking cycle.
- `[5, 0]`: **Builder Penguin** — Yellow construction hardhat.
- `[6, 0]`: **Support Tech Penguin** — Pink headset with boom microphone.
- `[7, 0]`: **Buyer / Dispatcher Penguin** — Logistics cap / formal apron.

Row 1 (Y: 64 – 127) — Workstations & PC Lifecycle

- `[0, 1]`: **Empty Build Desk** — Clean wooden workbench with metal legs and desk mat.
- `[1, 1]`: **Desk with Parts Box** — Unopened component boxes and GPU/RAM trays.
- `[2, 1]`: **Assembled PC** — Open mid-tower chassis seated on desk with keyboard.
- `[3, 1]`: **Installing Linux OS** — CRT/LCD monitor active with phosphor green terminal progress bar.
- `[4, 1]`: **Boxed & Labeled PC** — Sealed brown shipping carton with white barcode label.
- `[5, 1]`: **Admin Computer Station** — Dual-monitor sales desk with analytical charts.
- `[6, 1]`: **Helpdesk Station (Idle)** — Counter with vintage red rotary telephone and CRT monitor.
- `[7, 1]`: **Helpdesk Station (Active Alert)** — Flashing red siren light and vibrating telephone ringer.

Row 2 (Y: 128 – 191) — Environment & Logistics

- `[0, 2]`: **Garage Floor Tile** — 64×32 isometric concrete tile.
- `[1, 2]`: **Warehouse Floor Tile** — 64×32 polished wooden plank tile.
- `[2, 2]`: **Dispatch Loading Pad** — Yellow and black diagonal hazard warning floor pad.
- `[3, 2] & [4, 2]`: **Delivery Courier Van** — Isometric delivery truck with blue cargo bay.

Row 3 (Y: 192 – 255) — UI Icons & Badges

- `[0, 3]`: **Review Star Icon** — Gold 5-star rating icon.
- `[1, 3]`: **Shipping Carton Icon** — Small inventory icon.
- `[2, 3]`: **Terminal Prompt Icon** — Monospace `>_` icon.
- `[3, 3]`: **Cash Currency Badge** — Green circular dollar badge.

10. Room Backgrounds & Environment Architecture

10.1 Environment Design Guidelines

- **Cutaway Isometric Dihedral:** Rooms are constructed with two visible back walls (Northwest and Northeast) and an open foreground cutaway to maximize playability and prevent occlusion of player characters and workstations.
- **Aspect Ratio & Resolution:** 1280 × 720 px (native 16:9 720p HD canvas), scaling responsively in modern web viewports.
- **Tile Metrics:** 2:1 isometric diamond projection.
    - Garage Grid: 10 × 10 tiles (72px × 36px tile dimensions).
    - Warehouse Grid: 16 × 16 tiles (60px × 30px tile dimensions).

10.2 Garage Environment (Tier 1: Starter to 10 Desks)

- **Northwest Wall:**
    - Blue industrial roller shutter garage door opening directly out to the asphalt driveway and grass lawn where the courier van arrives.
    - High four-pane garage window allowing ambient daylight into the workshop.
- **Northeast Wall:**
    - Exposed brickwork lines and dark skirting board.
    - Workshop tool pegboard with wrenches, screwdrivers, and sticky notes.
    - Framed Linux Tux poster inspiring the penguin team.
- **Floor Plan & Designated Zones:**
    - Concrete slab floor with subtle alternating tile shading and grid alignment lines.
    - **Dispatch Bay (gy: 7–9, gx: 0–1):** Yellow and black diagonal hazard safety striping positioned directly beside the roller door for rapid box offloading.
    - **Admin Terminal (gy: 1–2, gx: 0–1):** Blue-accented floor zone near the back corner.
    - **Helpdesk Station (gy: 0–1, gx: 8–9):** Purple-tinted corner reserved for the support desk unlocked at Desk 10.
    - **10 Desk Slot Footprints (D1–D10):** Three rows of three desks plus one extra bay, marked with blue boundary outlines for clear spatial orientation.
- **Ambiance:** Dual hanging industrial fluorescent tube fixtures casting soft downlight cones across the assembly floor.

10.3 Warehouse Tier 1 (Tier 2: Up to 50 Desks & 5 Helpdesks)

- **Structural Elements:**
    - High corrugated steel paneling with dark vertical I-beams.
    - Overhead heavy industrial ventilation ducting running across the ceiling rafters.
    - Triple high-elevation industrial skylights along the upper Northeast wall.
- **Logistics & Departmental Layout:**
    - **Dual Shipping Bays (gy: 5–14, gx: 0–2):** Expanded hazard loading dock with two heavy-duty roller doors to accommodate higher van turnover.

    - **Customer Support Wing (gy: 0–4, gx: 12–15):** Dedicated department floor space for up to 5 concurrent Helpdesk stations.
    - **Main Assembly Hall (gy: 2–15, gx: 3–11):** High-density grid floor supporting scalable placement of up to 50 active build desks.

11. Penguin Animation System & Directional Perspectives

11.1 Isometric View Perspectives

To match the 2:1 isometric projection of the room, characters navigate across four primary isometric diagonals:

- **South-West (SW / Down-Left):** Front 3/4 perspective facing toward the bottom-left of the screen. Shows facial expressions, beak, eyes, and white chest belly.
- **South-East (SE / Down-Right):** Front 3/4 perspective facing toward the bottom-right.
- **North-West (NW / Up-Left):** Back 3/4 perspective facing away from the camera toward the top-left. Features back plumage, silhouette of beak, and rear foot lift.
- **North-East (NE / Up-Right):** Back 3/4 perspective facing away toward the top-right.

11.2 Animation Cycles & Frame Timing

- **Base Frame Rate:** 6.6 FPS (150 ms per frame) for walking, producing an exaggerated, cute penguin waddle.
- **Idle Cycle (2 Frames, 400 ms/frame):** Subtle breathing motion and resting flipper position.
- **Walk Cycle (4 Frames per Direction):**
    - *Frame 0 (Left Foot Plant):* Left webbed foot steps forward, right foot trailing, body tilts slightly.
    - *Frame 1 (Passing / Mid-Waddle):* Weight transfers, body bobs upward by 1–2 pixels, flippers counter-balance.
    - *Frame 2 (Right Foot Plant):* Right webbed foot steps forward, left foot trailing, opposite body lean.
    - *Frame 3 (Passing / Mid-Waddle):* Weight transfers back with upward bob.
- **Interacting / Assembling Cycle (2 Frames, 200 ms/frame):** Rapid flipper tapping and tool motion used during desk assembly, Linux installation, and admin listing.
- **Carrying Box Walk Cycle (4 Directions):** Penguin holds a sealed cardboard shipping box in both flippers at chest height while executing the 4-direction walk cycle to the dispatch placement bay.

11.3 Animation Sprite Sheet Atlas (512 × 512, 8×8 Grid)

- **Row 0:** Player 1 (Tux / Red Bowtie) — **South-West** (Idle: Col 0–1, Walk: Col 2–5, Interact: Col 6–7).
- **Row 1:** Player 1 (Tux / Red Bowtie) — **South-East** (Idle: Col 0–1, Walk: Col 2–5, Interact: Col 6–7).

- **Row 2:** Player 1 (Tux / Red Bowtie) — **North-West (Back)** (Idle: Col 0–1, Walk: Col 2–5, Interact: Col 6–7).
- **Row 3:** Player 1 (Tux / Red Bowtie) — **North-East (Back)** (Idle: Col 0–1, Walk: Col 2–5, Interact: Col 6–7).
- **Row 4:** Box-Carrying Walk Cycle across all 4 directions (SW: Col 0–1, SE: Col 2–3, NW: Col 4–5, NE: Col 6–7).
- **Row 5:** Player 2 (Co-op Helper / Blue Bandana) — SW and SE Walk Cycles (Col 0–7).
- **Row 6:** Player 2 (Co-op Helper / Blue Bandana) — NW and NE Back Walk Cycles (Col 0–7).
- **Row 7:** AI Worker Penguins (Yellow Construction Hardhat) across all 4 walking directions.